

A Closed Bounded Radius Counterexample for $c_0 \subset \ell_\infty$

Automated Math Discovery Smoke Test

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Source and Target

This packet concerns the closed-bounded version of Problem 1.1 discussed in:

Syamantak Das and Tanmoy Paul, *A study on various generalizations of Generalized centers (GC) in Banach spaces*, arXiv:2311.15818.

The paper answers the finite-subset version of Problem 1.1. In the observations section, the authors say that the corresponding question for closed bounded subsets is not known except in some trivial cases. The source location is page 6 of the paper.

1.3. Observations: In this subsection, we give an overview of our observations. Suppose that, Y has $n.X.I.P.$ in X and $F \in \mathcal{F}(Y)$, where $\text{card}(F) = n$ and X admits weighted Chebyshev centers for F . In section 2, it is demonstrated that Y admits restricted f -centers for F , where $\text{card}(F) = n$, for all continuous, monotone functions $f : \mathbb{R}_{\geq 0}^n \rightarrow \mathbb{R}_{\geq 0}$, if and only if for any collection of n closed balls with centers in F having a nonempty intersection in X also intersect in Y . We obtain a similar characterization to that [16, Theorem 2.7] for the family of closed and bounded subsets of Y .

Problem 1.1 is answered in Theorem 2.6, which extends a characterization to the notion $n.X.I.P.$ in Banach spaces. Consequently, we obtain various characterizations for central subspaces in Banach spaces, as stated in Theorem 2.7 and Theorem 2.8. It is easy to observe that if Y is a subspace of X and if there exists $P : X \rightarrow Y$ onto, where $P(\lambda x + y) = \lambda P(x) + y$, for $x \in X, y \in Y$ and scalar λ , $\|P(x)\| \leq \|x\|$ for all x , a *quasi-linear projection* $P : X \rightarrow X$, then for any $F \in \mathcal{B}(Y)$, $\text{rad}_Y(F) = \text{rad}_X(F)$. Vesely's example in [16, Pg.9] ensures there does not exist any quasi-linear projection $P : \ell_\infty \rightarrow c_0$. We do not know the solution to the question posed in Problem 1.1 for closed bounded subsets, with the exception of some trivial circumstances, such as when Y is 1-complemented in X or, more generally, a range of a quasi-linear projection. The requirement for Y being a central subspace can be more simply expressed when $Y \in (GC)$ is observed.

Let us give an example of a space X such that $L_1(\mu, X) \notin (GC)$. Let

Counterexample

Let $X = \ell_\infty$ and let $Y = c_0$ be its canonical closed subspace. For a bounded nonempty set A in a normed space E , write

$$\text{rad}_E(A) = \inf_{x \in E} \sup_{a \in A} \|x - a\|.$$

Theorem 1. *The pair $c_0 \subset \ell_\infty$ has*

$$\text{rad}_{c_0}(F) = \text{rad}_{\ell_\infty}(F)$$

for every finite subset $F \subset c_0$. Nevertheless there is a closed bounded subset $A \subset c_0$ such that

$$\text{rad}_{c_0}(A) \neq \text{rad}_{\ell_\infty}(A).$$

In fact one may take $A = \{e_n : n \in \mathbb{N}\}$, for which

$$\text{rad}_{\ell_\infty}(A) = \frac{1}{2}, \quad \text{rad}_{c_0}(A) = 1.$$

Proof. We first prove the finite-subset equality. The inequality $\text{rad}_{\ell_\infty}(F) \leq \text{rad}_{c_0}(F)$ is immediate because $c_0 \subset \ell_\infty$. Conversely, let $F \subset c_0$ be finite and let $r > \text{rad}_{\ell_\infty}(F)$. Choose $z \in \ell_\infty$ such that

$$\sup_{a \in F} \|z - a\|_\infty < r.$$

Since F is finite and each $a \in F$ tends to zero coordinatewise, there is N such that

$$\sup_{a \in F} \sup_{n > N} |a_n| < r.$$

Define $y \in c_0$ by

$$y_n = \begin{cases} z_n, & n \leq N, \\ 0, & n > N. \end{cases}$$

For $n \leq N$ the coordinate errors $|y_n - a_n|$ are the same as for z , and for $n > N$ they are $|a_n| < r$. Hence

$$\sup_{a \in F} \|y - a\|_\infty \leq r.$$

Letting $r \downarrow \text{rad}_{\ell_\infty}(F)$ gives $\text{rad}_{c_0}(F) \leq \text{rad}_{\ell_\infty}(F)$, and the finite-subset equality follows.

Now set $A = \{e_n : n \in \mathbb{N}\}$. This set is bounded in c_0 and is closed, since distinct elements of A are pairwise distance 1 apart.

In ℓ_∞ , the constant sequence $u = (1/2, 1/2, \dots)$ satisfies

$$\|u - e_n\|_\infty = \frac{1}{2} \quad (n \in \mathbb{N}),$$

so $\text{rad}_{\ell_\infty}(A) \leq 1/2$. For the reverse inequality, let $z \in \ell_\infty$ and suppose

$$\sup_n \|z - e_n\|_\infty \leq R.$$

Fix $k \in \mathbb{N}$. From the distance to e_k we get $|1 - z_k| \leq R$. From the distance to any e_m with $m \neq k$ we get $|z_k| \leq R$. Thus

$$R \geq \max\{|z_k|, |1 - z_k|\} \geq \frac{1}{2}.$$

Since k was arbitrary, $\text{rad}_{\ell_\infty}(A) \geq 1/2$, and therefore $\text{rad}_{\ell_\infty}(A) = 1/2$.

In c_0 , let $y \in c_0$. Since $y_n \rightarrow 0$,

$$\sup_n \|y - e_n\|_\infty \geq \sup_n |1 - y_n| \geq \limsup_n |1 - y_n| = 1.$$

Thus $\text{rad}_{c_0}(A) \geq 1$. The center $0 \in c_0$ gives $\sup_n \|0 - e_n\|_\infty = 1$, so $\text{rad}_{c_0}(A) = 1$. $\square$

Interpretation

The example shows that the finite criterion in Problem 1.1, and equivalently the finite intersection behavior captured by $n.X.I.P.$ for all n , does not control Chebyshev radii of arbitrary closed bounded subsets. The obstruction is not finite-dimensional: every finite part of $\{e_n\}$ has an ℓ_∞ -center that can be truncated into c_0 , but the limiting center $(1/2, 1/2, \dots)$ is not in c_0 .

Sanity Check Script

The file `code/check_radii.py` prints finite-prefix radius checks and illustrates how c_0 -like centers approach radius 1 on longer prefixes. The mathematical proof above is exact and does not depend on the script.